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Skill - 1

Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

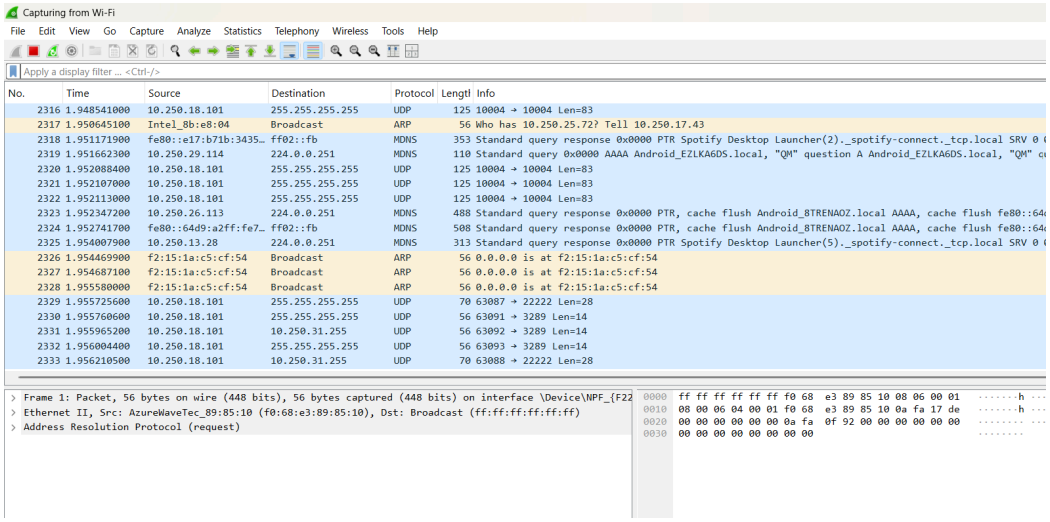
Step 1: Install Wireshark

1. Open Google Chrome.
2. Visit the official Wireshark download page: Under Windows Installer (64-bit), click the download link.
3. Wait for the installer to download.
4. Open the downloaded `.exe` file.
5. Follow the installation wizard:
    a. Click Next.
    b. Accept the license agreement.
    c. Keep the default components selected.
    d. Install Npcap when prompted.
    e. Complete the installation.



Step 2: Select your network interface

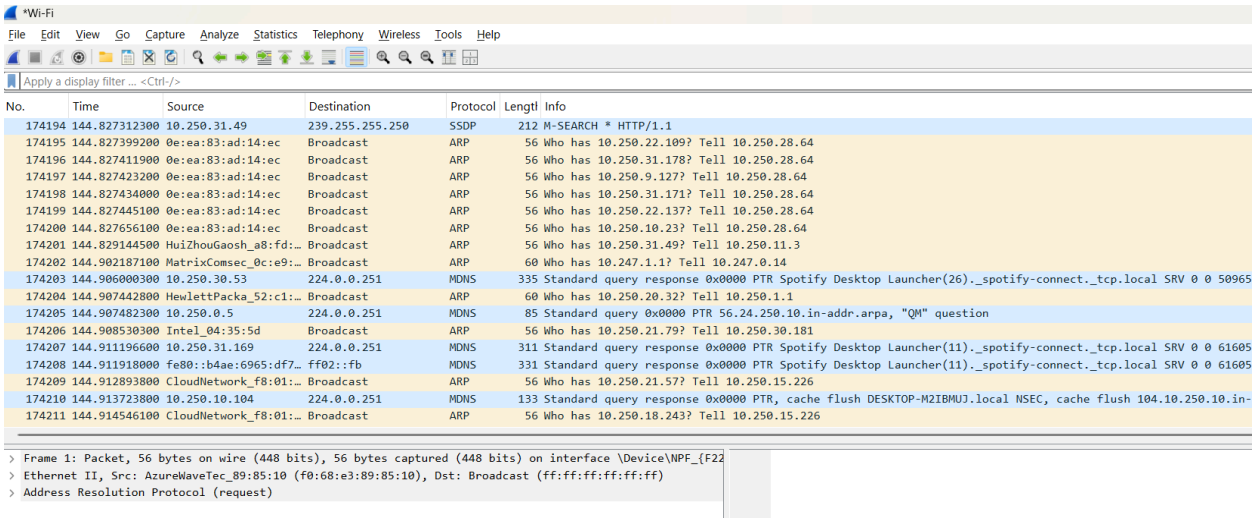
1. Open Wireshark.
2. Wait for the welcome screen to load.
3. Under the Capture section, you'll see available network interfaces, such as:
    - Wi-Fi
    - Ethernet
    - Bluetooth
    - Loopback
4. Identify the Wi-Fi interface (assuming your laptop is connected through Wi-Fi).
5. Double-click Wi-Fi.



Step 3: Generate network traffic

While Wireshark is capturing:

1. Open Google Chrome (don't close Wireshark).
2. Visit these websites one by one:
    - https://www.google.com
    - https://www.youtube.com
    - https://www.wikipedia.org
3. Browse each website for a few seconds.
4. Wait approximately 30 seconds in total.
5. Return to Wireshark.
6. Click the red Stop button at the top-left.

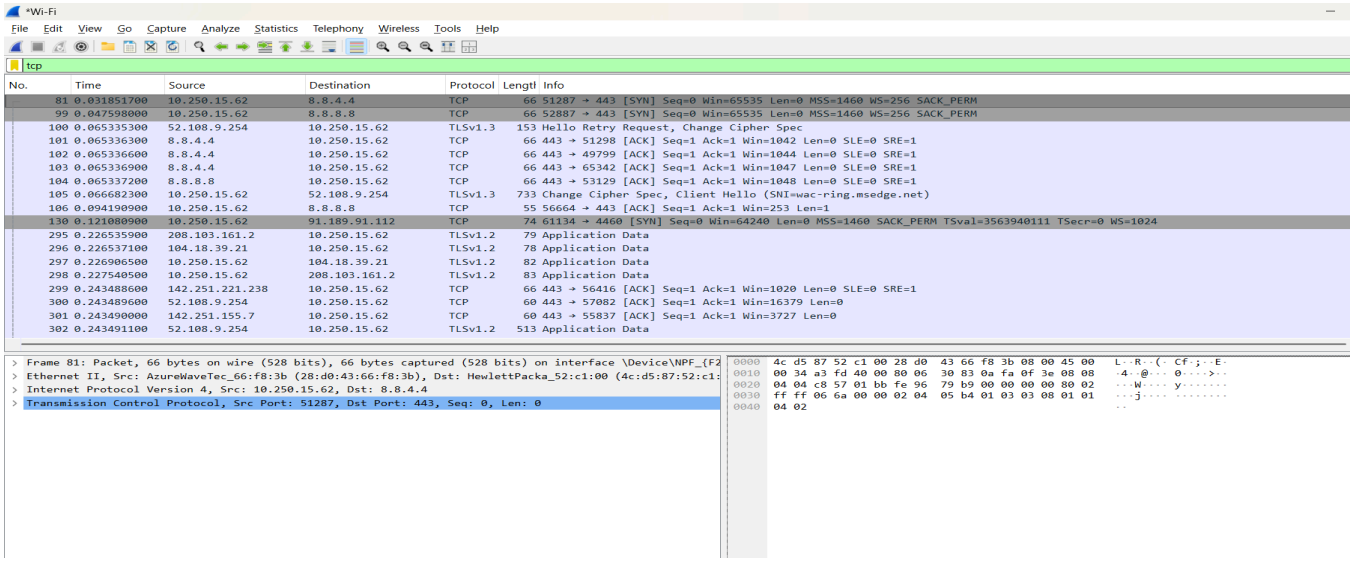


Step 4: Apply display filters

Now we'll identify TCP and UDP packets separately.

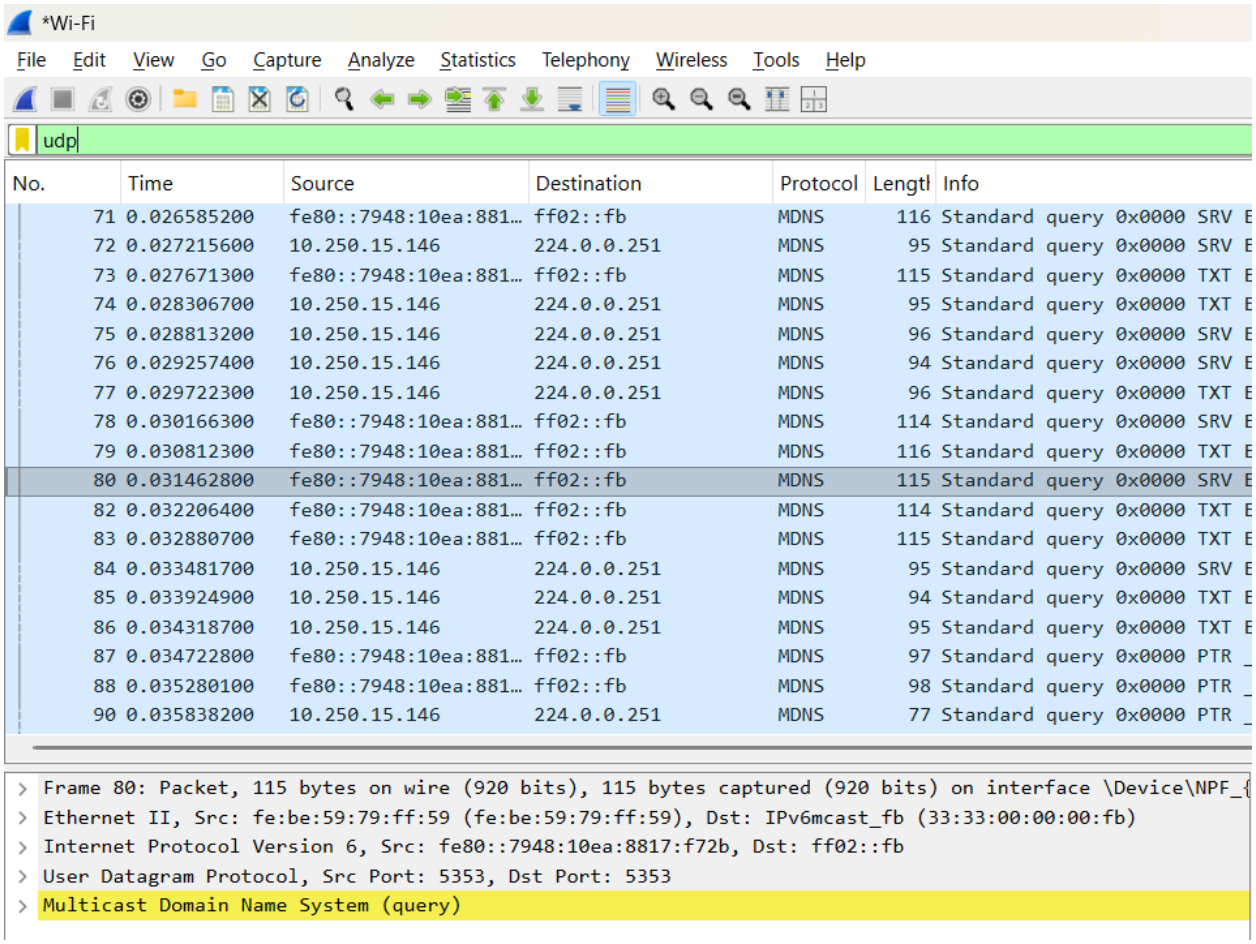
First: TCP packets

1. Locate the filter bar at the top of Wireshark.
2. Enter: tcp
3. Press Enter.
4. Wireshark will display only TCP packets.



Apply UDP filter

1. Click the display filter bar.
2. Delete `tcp`.
3. Type: udp
4. Press Enter.

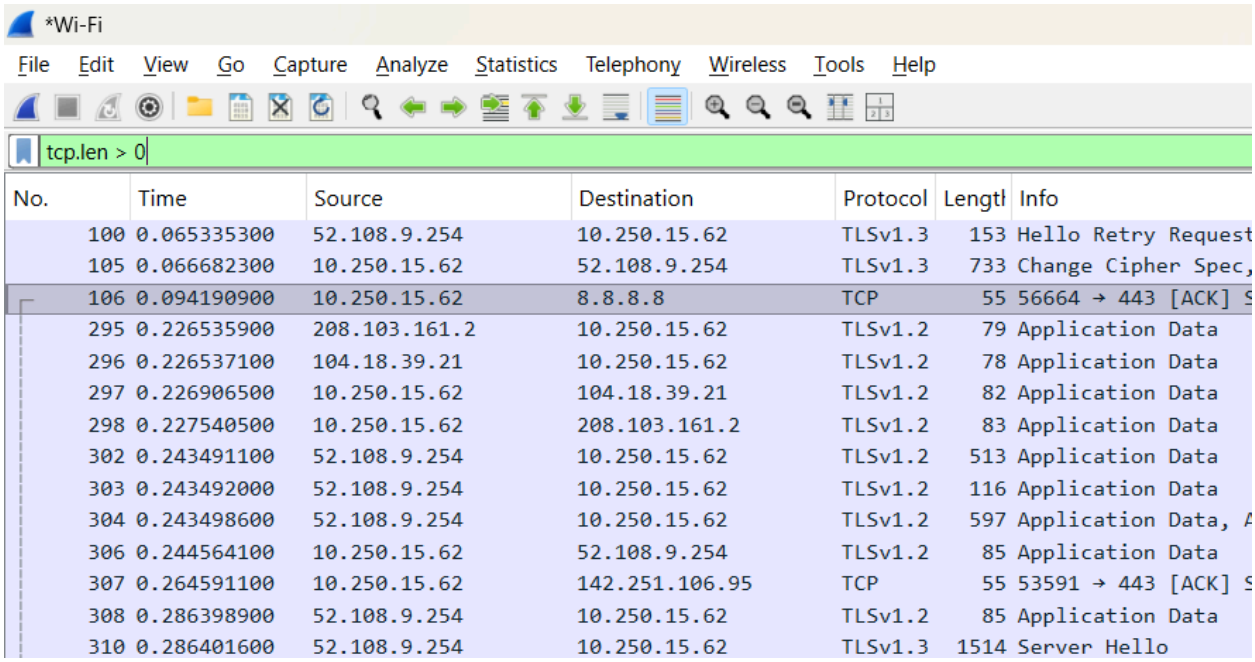


Step 5: Inspect packet headers

In this step, TCP and UDP packets were selected and their protocol headers were inspected using Wireshark.

A. TCP Packet Inspection

1. Applied the `tcp` display filter.
2. Selected packet number 310.



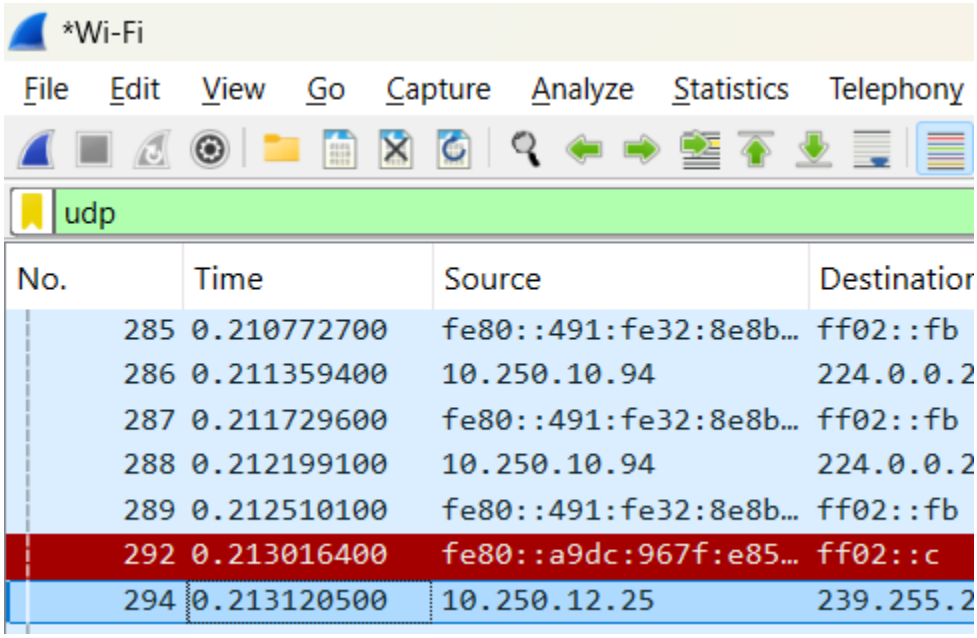
3. Expanded Frame, Ethernet II, IPv4 and TCP sections.
4. Recorded the following values:

Parameter	**Value**
Frame Length	1514 bytes
Ethernet Header	14 bytes
IPv4 Header	20 bytes
TCP Header	20 bytes
TCP Payload	1460 bytes

B. UDP Packet Inspection

1. Applied the udp display filter.

2. Selected packet number 294.



3. Expanded Frame, Ethernet II, IPv4 and UDP sections.
4. Recorded the following values:

Parameter	**Value**
Frame Length	169 bytes
Ethernet Header	14 bytes
IPv4 Header	20 bytes
UDP Header	8 bytes
UDP Payload	127 bytes

Step 6: Calculate Header Overhead

In this step, we calculate how many bytes are used by protocol headers and how much of the transmitted frame carries actual application data.

No additional Wireshark operation is required. We'll use the values collected in Step 5.

Part A: TCP Header Overhead

TCP packet (No. 310) contains:

Component	**Size**
Ethernet Header	14 bytes
IPv4 Header	20 bytes
TCP Header	20 bytes
TCP Payload	1460 bytes
Total Frame Length	1514 bytes

Total overhead:

$14 + 20 + 20 = 54 bytes$

Payload efficiency:

$$\frac{1460}{1514} \times 100 = 96.43\%$$

Part B: UDP Header Overhead

Your UDP packet (No. 294) contains:

Component	**Size**
Ethernet Header	14 bytes
IPv4 Header	20 bytes
UDP Header	8 bytes
UDP Payload	127 bytes
Total Frame Length	169 bytes

Total overhead:

$$14 + 20 + 8 = 42bytes$$

Payload efficiency:

$$\frac{127}{169} \times 100 = 75.15\%$$

Step 7: Fill in the Observation Table

In this step, we record the values obtained from Wireshark and compare TCP and UDP packet structures.

Observation Table

Parameter	TCP Packet (No. 310)	UDP Packet (No. 294)
Frame Length	1514 bytes	169 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	1500 bytes	155 bytes
Payload Length	1460 bytes	127 bytes

Observation

1. TCP has a larger transport header (20 bytes) than UDP (8 bytes).
2. TCP provides reliable data transmission, while UDP has lower header overhead.
3. The packet sizes and payload efficiencies differ depending on the amount of data transmitted.

Step 8: Save the Wireshark Capture File

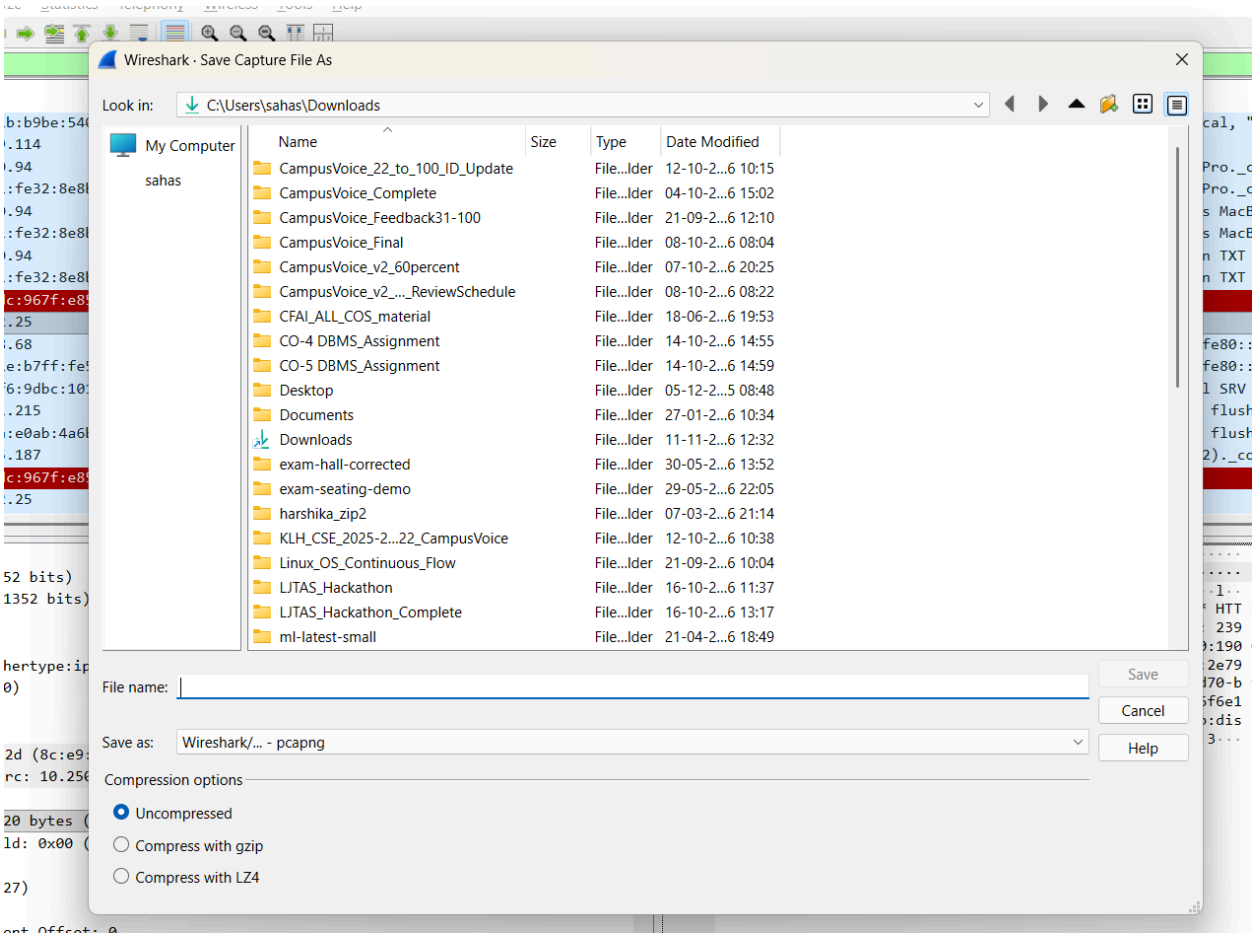
In this step, we save the captured network packets for future analysis.

Procedure

1. Make sure packet capturing is stopped.
2. Click File in the top menu.
3. Select Save As.
4. Choose a folder where you want to save the capture.
5. Enter the filename:

`Wireshark_Network_Capture`

6. Keep the file format as Wireshark/tcpdump - pcapng.
7. Click Save.



Step 9: Open a Saved Capture File and Apply ICMP Filter

In this step, we reopen the saved capture file and identify ICMP packets.

Procedure

1. Open Wireshark.
2. Click File → Open.
3. Navigate to the folder where you saved your `.pcapng` file.
4. Select your saved capture file.
5. Click Open.
6. In the display filter bar, enter:

icmp

7. Press Enter to view ICMP packets.

